

Journal Submission Forms and Declarations

Manuscript: “An Aharonov–Bohm Lattice Operator Framework for the Riemann ζ Zeros: A 38-Test Computational Verification Suite” · Author: Isaev Iskhak Khamzatovich (ORCID 0009-0003-7299-0701) · September 2026

1. CRediT author contributions

Isaev Iskhak Khamzatovich: Conceptualization; Methodology; Software (the AB-Cloud v23 suite, its full Julia implementation, and the Python reimplementations); Validation (design and execution of the 38-test verification run and all sub-check batteries); Formal analysis; Investigation; Data curation (the embedded 50,000-zero dataset, the run archive, and the repository record); Writing — original draft; Writing — review and editing; Visualization (the ABPlotV23 engine and all 60 figures); Project administration.

2. Declaration of interests

The author declares no competing interests, financial or non-financial, related to this work. The research was conducted independently; no institution, employer, or sponsor participated in the design, execution, or reporting of the study.

3. Funding statement

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. All computational resources were personally provisioned.

4. Data availability statement

All data, code, and run artefacts are openly available. Complete Julia source (v3.1): reproduced in Appendix B of the manuscript and archived at <https://github.com/wild8highlander/ab-cloud-research>. Run archive (run_20260914_234626): 38 test folders with reports, logs, 60 figures, and 36 animations; archived with DOI 10.5281/zenodo.21825394. Python reimplementations: shipped with the submission package. There are no restrictions on availability; no data were collected from human or animal subjects.

5. Ethics and consent

Not applicable: the study is computational and involves no human participants, animals, or personal data.

6. Author summary (for editorial triage)

The Hilbert–Pólya programme seeks a quantum system whose spectrum reproduces the zeros of the Riemann zeta function. This work presents an Aharonov–Bohm lattice construction proposed for that role and subjects it to a thirty-eight-test adversarial verification: exact gauge and symmetry certificates, GUE-class spectral statistics, topological anchors, Dirac-cone universality, and analytic constants — with all acceptance windows pre-registered and all thirteen failing verdicts analysed rather than hidden. The full source and run archive are open, making the study completely reproducible.

7. Suggested reviewers (optional, to be completed by the author)

To be completed by the author at submission time. The manuscript's subject area spans computational physics (random-matrix theory), analytic number theory (zeta statistics), and topological condensed-matter physics (Chern insulators, Dirac systems, Hatano–Nelson deformation); reviewers from any of these communities with random-matrix-computation experience are appropriate.

8. Licence

The author licences the work for publication under the journal's open licence (CC BY 4.0 recommended for OA venues) and releases the accompanying code under an MIT-style licence as recorded in the repository.